

SUMMARY

Early-career C++ developer focused on Linux, systems programming, and real-time applications. Building projects in C++17 on Unix-like environments and learning networking and embedded concepts. Seeking a C/C++ working-student role to grow in industrial automation, debugging, and modern software engineering practices.

TECHNICAL SKILLS

C++(intermediate) | OOP / OOD | STL | Boost(familiar) |
CMake(basic) | Unit testing (learning) | Perforce | Git |
UMTS/LTE (basic) | Linux(basics)

PROFESSIONAL SKILLS

Problem solving | Debugging | Team collaboration | Jira
Agile teamwork & sprint delivery | Time management |
Technical communication | Project ownership

PROJECTS

Additional Skills

TG_3SIXO — Real-Time Traffic Simulation Backend

Sep 2025 - Oct 2025

C++ | JS/HTML | CMake | FastAPI | HackYeah

- Built a real-time C++17 routing and prediction engine modeled after telecom-grade network routing.
- Implemented an event-driven backend with SSE, multithreading, and thread-safe data structures.
- Designed custom Dijkstra-based routing with dynamic weight updates similar to link-state recalculations.
- Developed Transit Engine, a lightweight C++ analytics module for pattern detection and delay forecasting.

C++ Game Engine — Custom rendering pipeline (in progress).

Nov 2025 - Current

C++17 | OpenGL | GLSL | GLM | GLFW | ImGui | Linux

- Developing a modular game engine featuring a renderer, scene manager, shader system, and simple ECS.
- Implemented vertex/element buffers, shader abstraction, perspective projection, entity rendering, and transform logic.
- Rendering-textured 3D objects (triangle/cube) with camera controls and UI integration using ImGui.

CLI To-Do List — Embedded-Style Command-Line Utility.

Aug 2025 - Sep 2025

C++17 | CMake | Linux | File System | State Management

- Built a deterministic command-line tool with a custom REPL loop and similar to embedded diagnostic CLIs.
- Designed a Task class with encapsulated state, auto-increment IDs, and safe text-file persistence and an update logic.
- Followed industry tooling and structure (CMake, modular src/ + include/, Git) to mirror real embedded workflows.

Additional Projects

EDUCATION

Bachelor of Engineering in Computer Science

Aug 2020 - Jun 2024

Chandigarh University

Diploma (Programming Specialization in Game Engine dev.)

Sep 2024 - Dec 2026

FutureGames

ADDITIONAL INFORMATION

- Languages:** English (Fluent) , Hindi (Native) , Gujarati (Native)
- Certifications:** Intermediate OOPS for Unity Games , Object-Oriented Data Structures in C++ , Cloud Virtualization, Containers and APIs
- Awards/Activities:** Organized a Hackathon in 2022 , Participated In Hackyeah2025

References

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